Tutorial Sheet-05

1. (a) Let (X, Y) have joint density f. Show that X and Y are independent if and only if for some constant $k > 0$ and nonnegative functions f_1 and f_2,

$$f(x, y) = kf_1(x)f_2(y)$$

for all $x, y \in \mathbb{R}$.

(b) Let $A = \{x : f_X(x) > 0\}$ and $B = \{y : f_Y(y) > 0\}$, where f_X and f_Y are the marginal densities of X and Y, respectively. Show that if X and Y are independent, then

$$\{(x, y) : f(x, y) > 0\} = A \times B.$$

2. Let the joint PDF of (X, Y) be given by $f(x, y) = 2$ on the triangular region $D = \{(x, y) : 0 < x < y < 1\}$ and 0 otherwise. Find the conditional probability density function $f_{X|Y}(x|y)$ using the marginal PDF of Y.

3. Let (X, Y) be uniformly distributed over the square region rotated by 45°, defined by $|x|+|y| \leq 1$. The joint PDF is $f(x, y) = 1/2$ inside this region and 0 outside.

   (a) Find the marginal PDFs $f_X(x)$ and $f_Y(y)$.

   (b) Check if X and Y are independent using the factorization condition $f(x, y) = f_X(x)f_Y(y)$.

   (c) Explain why independence fails here even though the joint PDF $f(x, y)$ appears to be a constant.

4. Consider three random variables X_1, X_2, X_3 defined as follows: X_1 and X_2 are independent coin tosses, taking values $\{0, 1\}$ with probability 0.5 each. Define $X_3 = |X_1 - X_2|$ (equivalent to X_1 XOR X_2) (XOR stands for Exclusive OR).

   (a) Show that the pair (X_1, X_3) is independent.

   (b) Show that the pair (X_2, X_3) is independent.

   (c) Verify whether the triplet (X_1, X_2, X_3) is mutually independent by checking the joint condition:

$$P(X_1 = x, X_2 = y, X_3 = z) = P(X_1 = x)P(X_2 = y)P(X_3 = z).$$

5. Let X and Y be independent and identically distributed (i.i.d.) random variables with common probability density function (PDF):

$$f(x) = \begin{cases} 1, & 0 < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Find the PDFs of the following random variables:

$$XY, \quad \frac{X}{Y}, \quad \min\{X, Y\}, \quad \max\{X, Y\}$$

6. Let X_1, X_2, X_3 be i.i.d. random variables with common density function:

$$f(x) = \begin{cases} 1, & 0 < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Show that the PDF of $U = X_1 + X_2 + X_3$ is given by:

$$g(u) = \begin{cases} \dfrac{u^2}{2}, & 0 < u < 1, \\ 3u - u^2 - \dfrac{3}{2}, & 1 < u < 2, \\ \dfrac{(u-3)^2}{2}, & 2 < u < 3, \\ 0, & \text{elsewhere.} \end{cases}$$

(An extension to the n-variate case holds).

7. Let (X, Y) be distributed with joint density:

$$f(x,y) = \begin{cases} \frac{1}{4}\left[1 + xy(x^2 - y^2)\right], & \text{if } |x| \le 1, |y| \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

Find the Moment Generating Function (MGF) of (X, Y). Are X and Y independent? If not, find the covariance between X and Y.

8. Let X_1, X_2, X_3 be independent and identically distributed (i.i.d.) random variables with the common standard normal probability density function (PDF):

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}, \quad -\infty < x < \infty.$$

Consider the transformation given by:

$$Y_1 = \frac{X_1 - X_2}{\sqrt{2}},$$
$$Y_2 = \frac{X_1 + X_2 - 2X_3}{\sqrt{6}},$$
$$Y_3 = \frac{X_1 + X_2 + X_3}{\sqrt{3}}.$$

(a) Find the Jacobian of the transformation from (X_1, X_2, X_3) to (Y_1, Y_2, Y_3).

(b) Verify that $X_1^2 + X_2^2 + X_3^2 = Y_1^2 + Y_2^2 + Y_3^2$.

(c) Show that the joint PDF of (Y_1, Y_2, Y_3) is given by:

$$h(y_1, y_2, y_3) = \frac{1}{(\sqrt{2\pi})^3} \exp\left(-\frac{y_1^2 + y_2^2 + y_3^2}{2}\right).$$

(d) Conclude whether Y_1, Y_2, Y_3 are also independent R.V.